

Photosynthesis

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus Photosynthesis uses sunlight, carbon dioxide, and water to produce glucose and oxygen:
 $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.

Round 1 - Retrieve It

1. What are the three main inputs for photosynthesis?

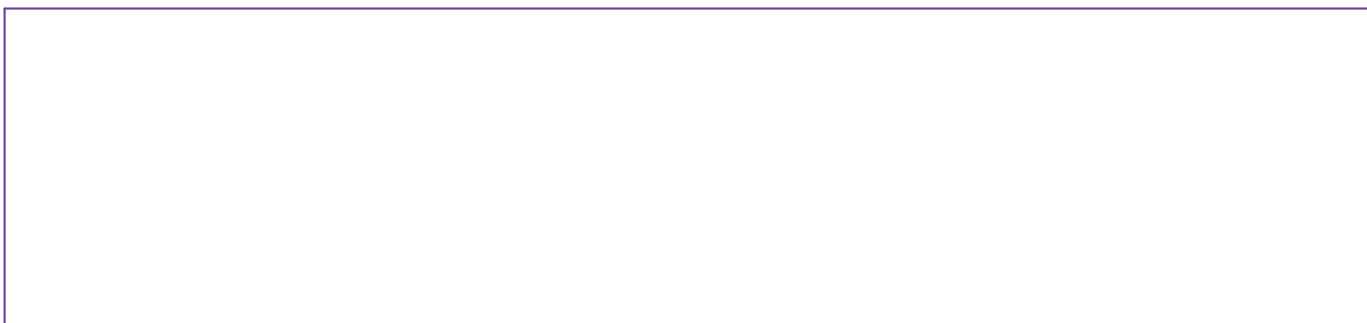
2. What are the main products?

3. In which organelle does photosynthesis occur in plant cells?

4. What form of stored chemical energy is produced?

Round 2 - Sketch It

Draw a leaf with arrows showing sunlight, CO_2 , and H_2O going in and glucose and O_2 coming out. Label the chloroplast as the cellular site.



Photosynthesis - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. Count atoms on each side of $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. Is the equation balanced?

2. A plant is kept in darkness but still has water and CO_2 . Which required input is missing?

3. Why is glucose important to the plant?

Round 4 - Reason from Evidence

Explain how photosynthesis connects energy from the Sun to consumers in a food web.

Photosynthesis - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Photosynthesis in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect photosynthesis to gas exchange. Which gas is taken in and which is released during photosynthesis?

E**Extend**

Predict how reducing available light for several days could affect a plant's glucose production, assuming other factors stay similar.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Photosynthesis

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Light energy, carbon dioxide, and water.
2. Glucose and oxygen.
3. Chloroplast.
4. Glucose/sugar.

Sketch It - look for

Inputs and outputs oriented correctly; chloroplast identified; glucose shown as product/stored chemical energy.

Apply It

1. Yes: C = 6, H = 12, O = 18 on both sides.
2. Light energy.
3. It stores chemical energy and provides carbon/building material for growth and metabolism.

Reason from Evidence - look for

Producers capture light energy and store it in glucose/biomass; consumers obtain chemical energy by eating producers or organisms that ate producers.

ACE - Extend look-for

Glucose production should decrease because light energy is required to drive photosynthesis.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.